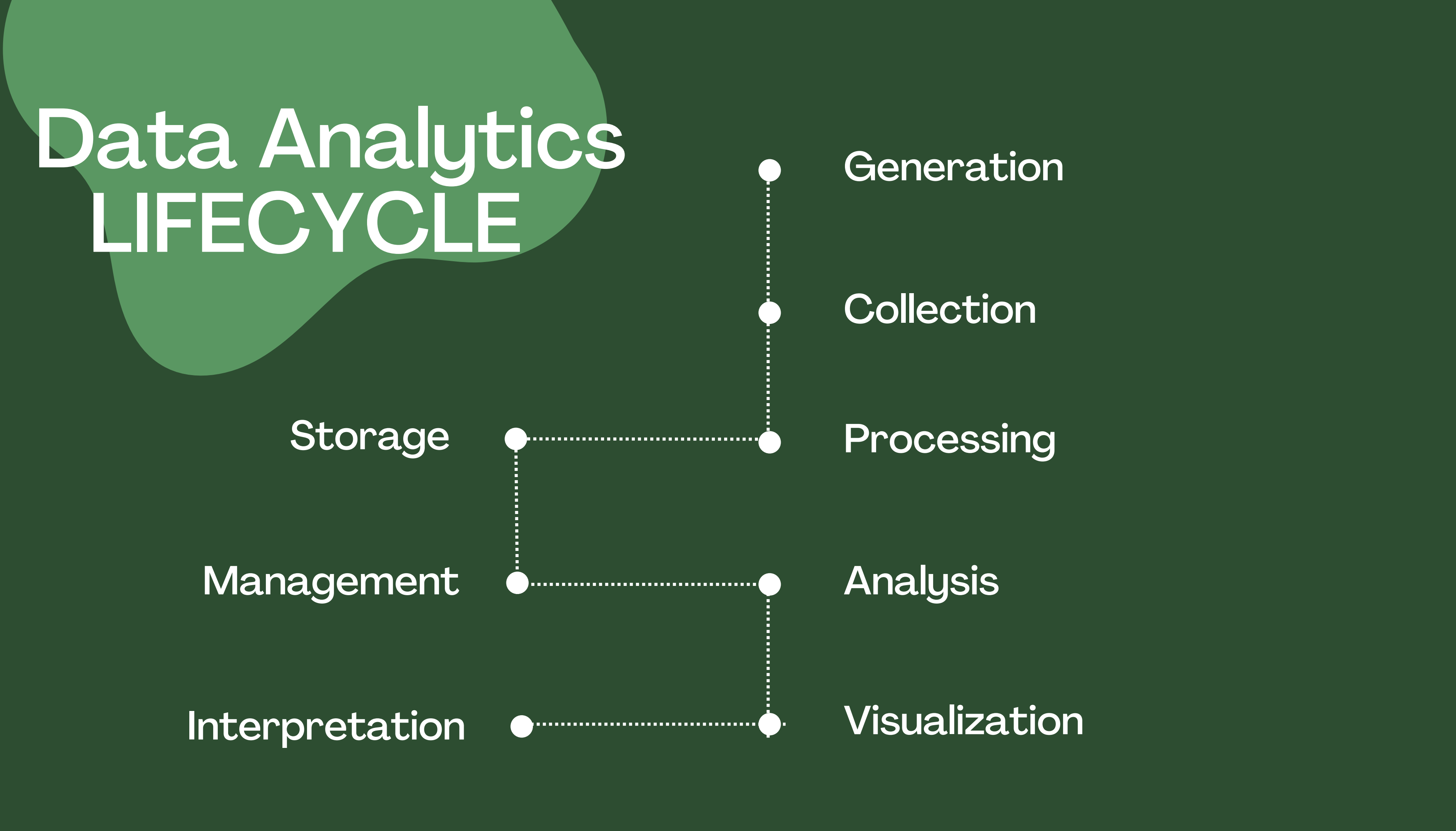


Texas Tribune — Government Employee Salaries



Presented by:
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Data Analytics LIFECYCLE



Problem Statement

“ Does salary equity exist across gender and ethnic groups in the Texas state government ? ”

—

What are those disparities and which group are affected ?

Generation & Collection

Source & variables

- Dataset from the Texas Tribune GOV Salaries Explorer.
- Contains salary information for more than 150,000 Texas GOV employees.
- Variables: name, monthly salary, yearly salary, gender, ethnicity...

Collection process

- Tribune collects the data directly from the Texas Comptroller of Public Accounts.
- The salary information is obtained through formal public information requests filed under the Texas Public Information Act (TPIA)

Application

- Allow transparencies
- Allow policymakers and researchers to notice patterns of pay discrepancies
- Agencies can make data-driven decisions to better compensation in Texas

Processing with Python

- Removed irrelevant columns (class title, state code, summed rate, etc)
- Combined and standardized first, middle, and last names
- Fixed dates and created a tenured column
- Removed duplicates

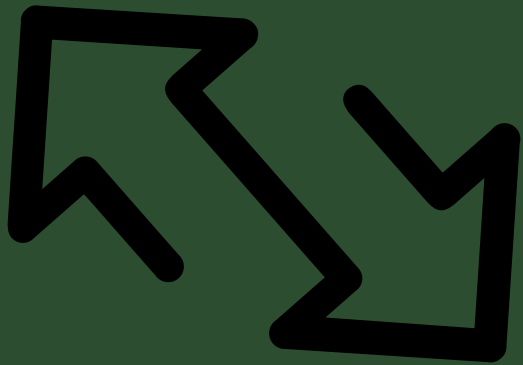
Processing with Python

- Normalized text & Numerized salaries
- Standardize ethnicity
- Extracted employee type

Processing

	AGENCY	AGENCY NAME	LAST NAME	FIRST NAME	MI	CLASS CODE	CLASS TITLE	ETHNICITY	GENDER	EMPLOYEE TYPE	...	RATE	HRSWKD	MONTHLY	ANNUAL	STATENUM	duplicated	multiple
0	529	HEALTH AND HUMAN SERVICES COMMISSION ...	AMINU	MUJAHID		5620	ELIGIBILITY ADVISOR I ...	BLACK	MALE	CTF - CLASSIFIED TEMPORARY FULL-TIME	...	0.00000	40.0	3075.10	36901.20	1732214	True	
1	320	TEXAS WORKFORCE COMMISSION ...	BARBER	BRITTANY		5062	VOC REHAB CNSLR I ...	BLACK	FEMALE	CRF - CLASSIFIED REGULAR FULL-TIME	...	0.00000	40.0	4477.00	53724.00	993461	True	
2	101	SENATE ...	CABALLERO	KARINA	G	7103	LEG. SERVICE/MAINTENANCE ...	HISPANIC	FEMALE	URF - UNCLASSIFIED REGULAR FULL-TIME	...	0.00000	41.0	4408.33	52899.96	1746417	True	
3	530	DEPARTMENT OF FAMILY AND PROTECTIVE SERVICES ...	CARDENAS	JACQUELINE	M	1688	MANAGEMENT ANALYST IV ...	HISPANIC	FEMALE	CRF - CLASSIFIED REGULAR FULL-TIME	...	0.00000	40.0	7944.09	95329.08	139905	True	
4	241	COMPTROLLER'S - JUDICIARY ...	CARR	BRENT	A	CC23	RETIRED COUNTY COURT JUDGE ...	WHITE	MALE	URP - UNCLASSIFIED REGULAR PART-TIME	...	96.15381	29.0	12083.32	144999.84	1634399	True	

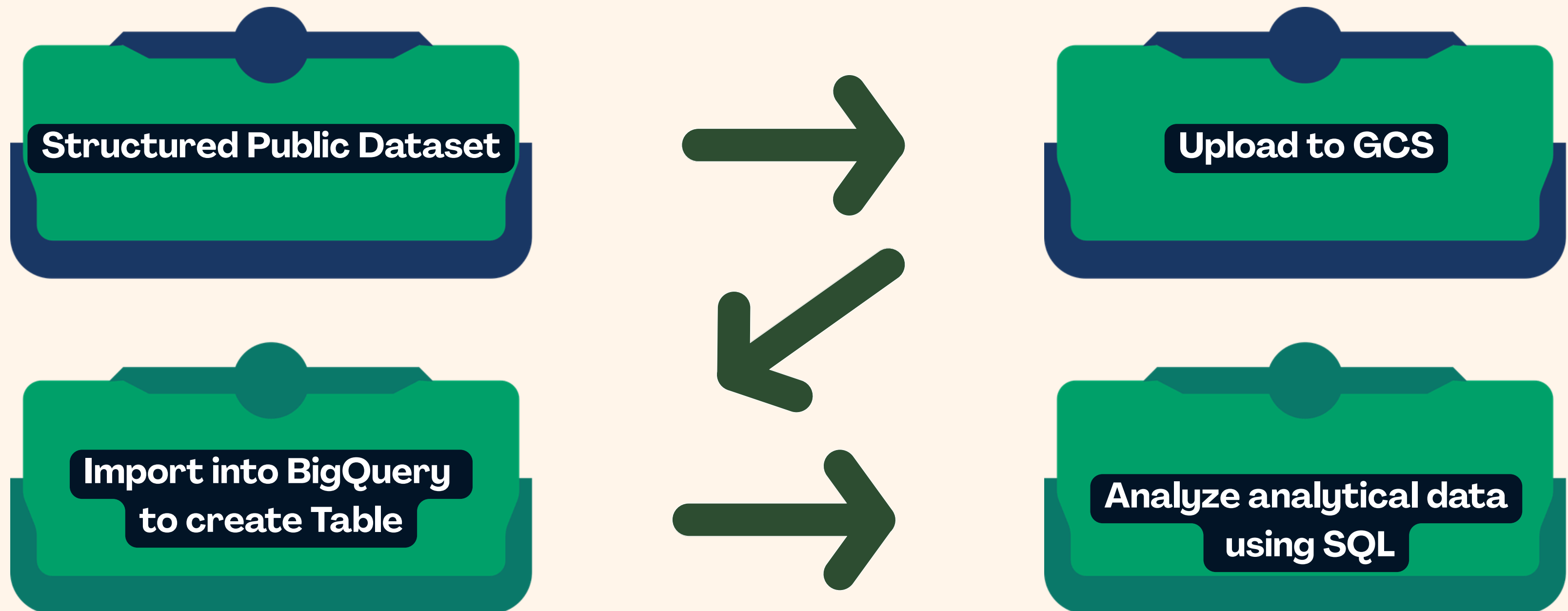
Before



After

...	AGENCY	AGENCY NAME	FULL_NAME	CLASS TITLE	EMPLOYEE TYPE	ETHNICITY	GENDER	HIRE DATE	TENURE_YEARS	RATE	HRSWKD	MONTHLY	ANNUAL	summed_annual_salary
6919	529	HEALTH AND HUMAN SERVICES COMMISSION	NORMA J ARREOLA	CLERK III	CRF - CLASSIFIED REGULAR FULL-TIME	HISPANIC	FEMALE	2004-09-01	21.271320	0.0	40.0	3885.53	46626.36	NaN
21588	529	HEALTH AND HUMAN SERVICES COMMISSION	MONICA CARGILL	ELIGIBILITY ADVISOR II	CRF - CLASSIFIED REGULAR FULL-TIME	HISPANIC	FEMALE	2019-09-30	6.183649	0.0	40.0	3790.35	45484.20	NaN
53247	696	TEXAS DEPARTMENT OF CRIMINAL JUSTICE	JOSEPH K GRAY	CORRECTIONAL OFFICER IV	CRF - CLASSIFIED REGULAR FULL-TIME	WHITE	MALE	2016-07-05	9.422005	0.0	40.0	5141.24	61694.88	NaN
118183	405	DEPARTMENT OF PUBLIC SAFETY	CESAR RODRIGUEZ	TROOPER	CRF - CLASSIFIED REGULAR FULL-TIME	HISPANIC	MALE	2015-07-12	10.405566	0.0	40.0	7165.75	85989.00	NaN
89524	696	TEXAS DEPARTMENT OF CRIMINAL JUSTICE	CHRISTAL U MEADOR	CORRECTIONAL OFFICER IV	CRF - CLASSIFIED REGULAR FULL-TIME	BLACK	FEMALE	2025-01-29	0.846662	0.0	40.0	5256.02	63072.24	NaN
110518	529	HEALTH AND HUMAN SERVICES COMMISSION	ANN-MARIE PRICE	PROJECT MANAGER IV	CRF - CLASSIFIED REGULAR FULL-TIME	WHITE	FEMALE	2024-08-28	1.268580	0.0	40.0	8347.50	100170.00	NaN

Data Storage



Data Analysis

Is there a pay gap between genders? Is it significant ?

The screenshot shows a web-based data analysis tool interface. At the top, there's a search bar and a navigation menu. Below the navigation menu, there's a toolbar with buttons for 'Run', 'Save', 'Download', 'Share', 'Schedule', 'Open in', and 'More'. The main area displays a SQL query for an 'Untitled query'.

```
1 SELECT
2   GENDER,
3   AVG(ANNUAL) AS avg_salary,
4   APPROX_QUANTILES(ANNUAL, 100)[SAFE_OFFSET(50)] AS median_salary,
5   COUNT(*) AS employee_count
6 FROM `dieng4240.Texasdataset.Texasclean`
```

Below the query, a status bar indicates 'Query completed' and 'Using on-demand processing quota'. The 'Query results' section shows a table with the following data:

Row	GENDER	avg_salary	median_salary	employee_count
1	MALE	73132.59052004...	63072.24	65418
2	FEMALE	65060.23001485...	58552.56	87506

Data Analysis

Are ethnic groups compensated the same for same job?

DIENG4240

Search (/) for resources, docs, products, and more

Search

1

Untitled query

Run

Save

Download

Share

Schedule

Open in

More

```
1 SELECT
2   CLASS_TITLE,
3   ETHNICITY,
4   AVG(ANNUAL) AS avg_salary,
5   COUNT(*) AS employee_count
6 FROM `dieng4240.Texasdataset.Texasclean`
```

Query completed

Using on-demand processing quota

Query results

Save results

Open in

Job information

Results

Visualization

JSON

Execution details

Execution graph

Row	CLASS_TITLE	ETHNICITY	avg_salary	employee_count
1	ACCOUNTANT I	HISPANIC	51711.13636363...	44
2	ACCOUNTANT I	WHITE	51144.87999999...	39
3	ACCOUNTANT I	BLACK	49529.66857142...	21

Data Analysis

Is a certain demographic paid more hourly despite same tenure?

DIENG4240

Search (/) for resources, docs, products, and more

Search

1

>|

diengbasic

Texasclean

*Untitled...ery

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+

Untitled query

Run

Save

Download

Share

Schedule

Open in

More

```
1 SELECT
2   ETHNICITY,
3   GENDER,
4   AVG(ANNUAL / (HRSWKD * 52)) AS avg_hourly_wage,
5   AVG(HRSWKD) AS avg_hours,
6   COUNT(*) AS employee_count
```

Query completed

Using on-demand processing quota

Query results

Save results

Open in

Job information

Results

Visualization

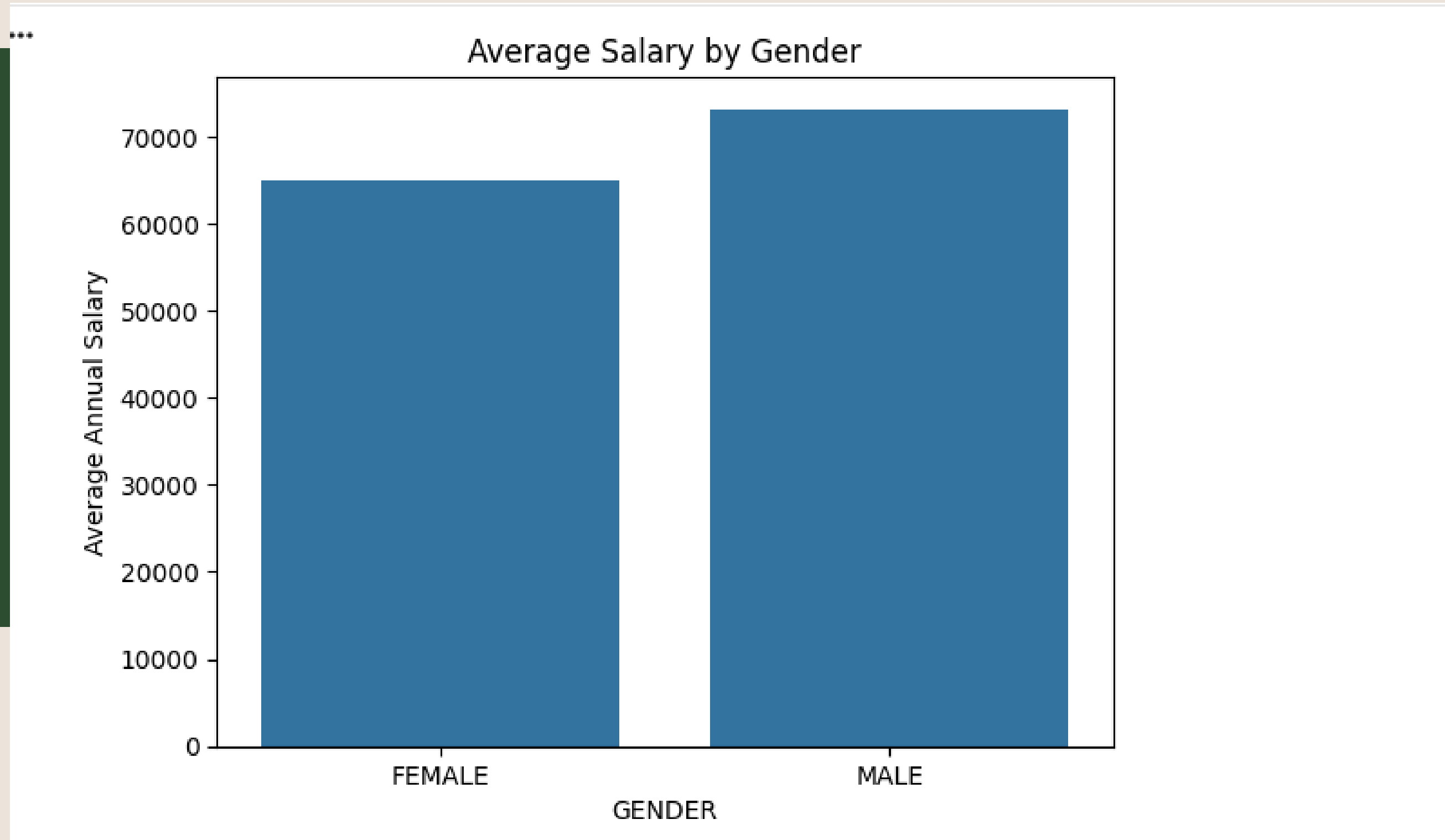
JSON

Execution details

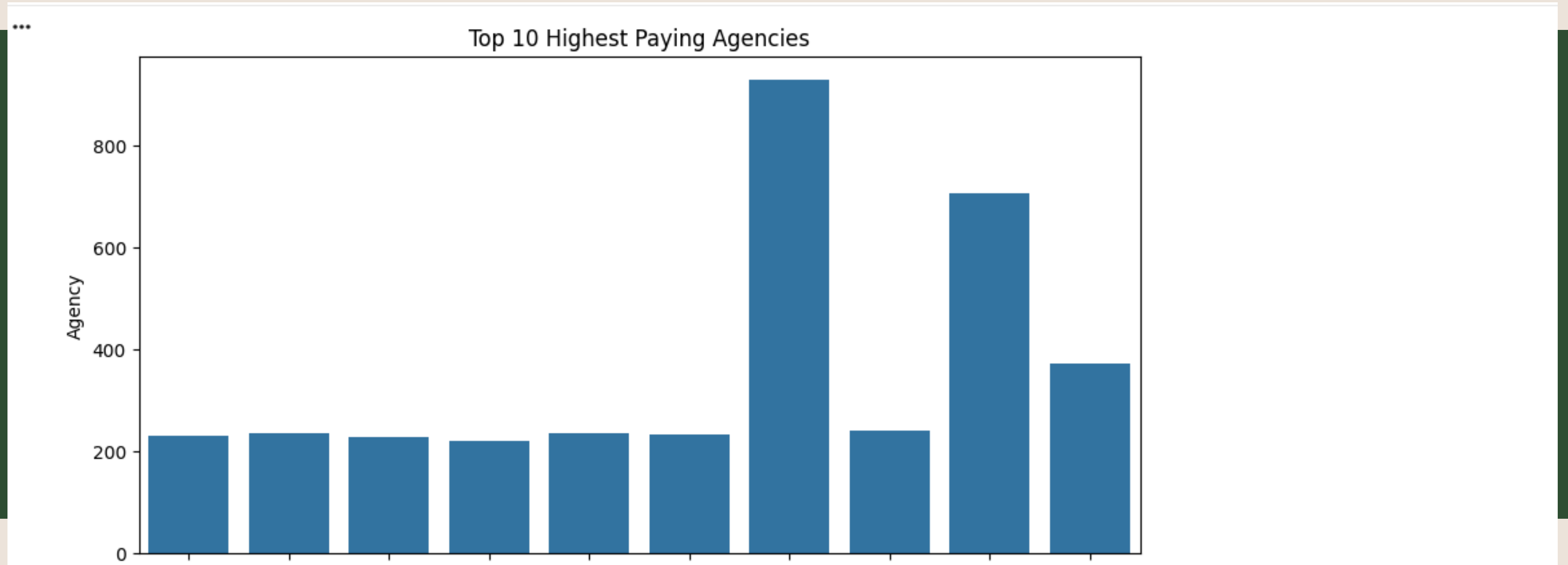
Execution graph

Row	ETHNICITY	GENDER	avg_hourly_wage	avg_hours	employee_count
1	ASIAN	MALE	42.78737467350...	39.62484103433...	2359
2	ASIAN	FEMALE	40.42379588942...	39.61197321136...	3061
3	WHITE	MALE	39.18581646994...	39.55680948252...	29992

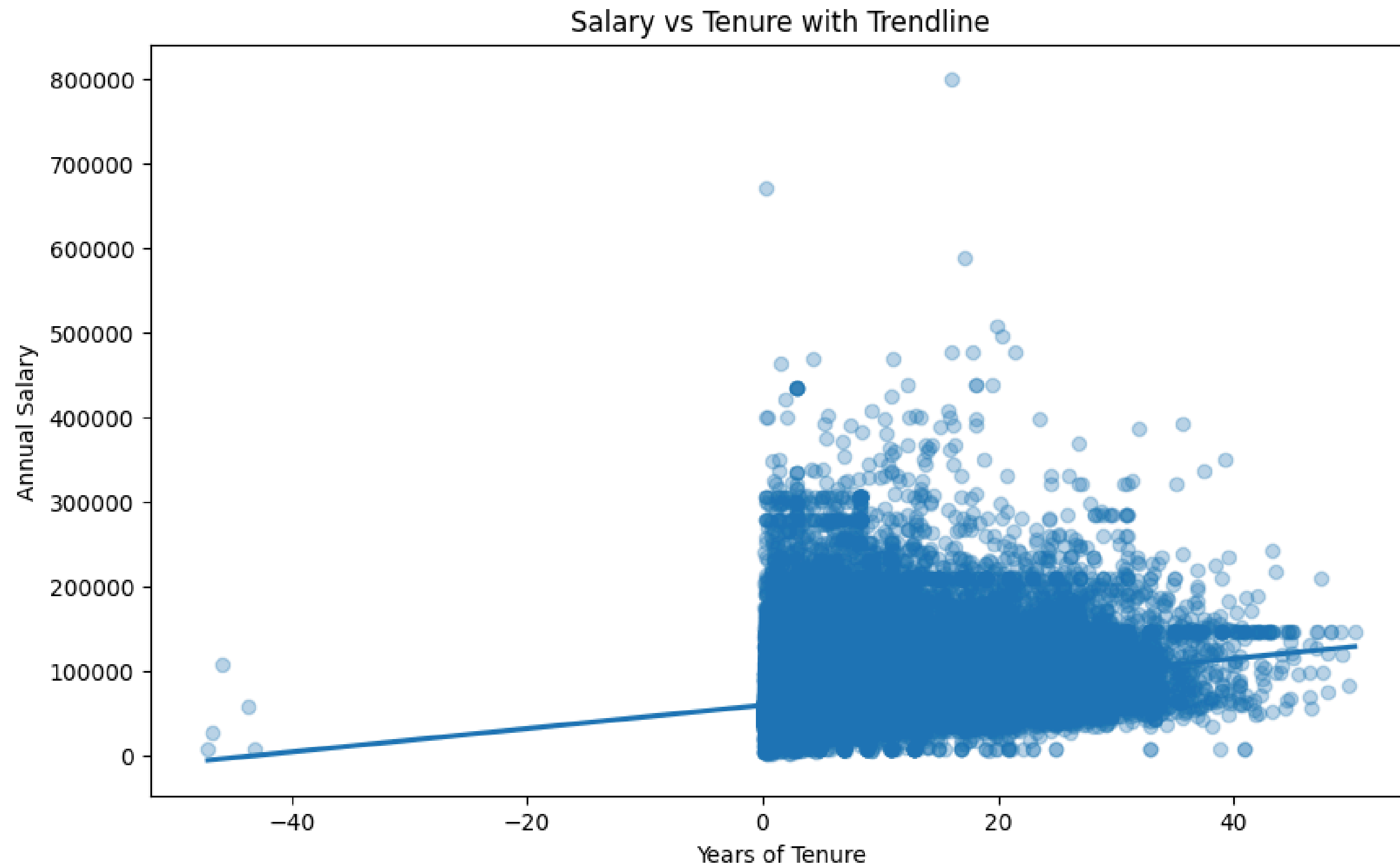
Visualization with python



Visualization with python



Visualization with python



Interpretation



12%

Gender pay gap favoring men

Black employees

\$2,200 LESS

\$30,000 GAP

Black Women lowest earning group.

Thank you
Questions?



References

Texas Tribune. (n.d.). Texas government salaries.
Retrieved October 1, 2025, from
<https://salaries.texastribune.org/>